

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	32.8	8	1.5	0.3	0.3	57.1
U3 - Toxicity prior Beta(0.15, 0.85)	34.7	7.4	0.5	0.2	0	57.2
U2 - Toxicity prior Beta(0.3, 0.7)	33.7	8.3	1.6	0.7	0.2	55.5
U3 - Toxicity prior Beta(0.3, 0.7)	37.3	6.5	0.7	0.2	0.1	55.2
U2 - Toxicity prior Beta(0.5, 0.5)	32.4	10.9	1.4	0.8	0.3	54.2
U3 - Toxicity prior Beta(0.5, 0.5)	37.8	6.4	1.1	0.2	0.2	54.3
U2 - Toxicity prior Beta(1, 1)	34.7	8.9	1.3	0.2	0.3	54.6
U3 - Toxicity prior Beta(1, 1)	36.3	7.8	1	0.2	0.1	54.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15.1	4.2	2.1	0.8	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	15.2	4.1	2.1	0.8	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	14.8	4.2	2.4	1	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	14.7	4.3	2.4	1	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	14.6	4.1	2.6	1.3	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	14.5	4.1	2.6	1.3	0.5	
U2 - Toxicity prior Beta(1, 1)	14.5	4.1	2.6	1.3	0.6	
U3 - Toxicity prior Beta(1, 1)	14.5	4.2	2.6	1.3	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	1.9	0.4	0.2	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	1.9	0.4	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	1.7	0.4	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	1.7	0.4	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	1.6	0.3	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	1.6	0.3	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.4	1.4	0.3	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.4	1.4	0.3	0.2	0	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	36.5	19.5	29.6	10.3	2.2	1.9
U3 - Toxicity prior Beta(0.15, 0.85)	39.5	21.9	26.1	8.7	1.9	1.9
U2 - Toxicity prior Beta(0.3, 0.7)	35.1	17.8	30.2	12.5	2.6	1.8
U3 - Toxicity prior Beta(0.3, 0.7)	37.5	19.4	28.6	10.4	2.2	1.9
U2 - Toxicity prior Beta(0.5, 0.5)	35.5	16.9	29.6	12.8	3.3	1.9
U3 - Toxicity prior Beta(0.5, 0.5)	37.5	18.7	27.5	12.2	2.2	1.9
U2 - Toxicity prior Beta(1, 1)	34.2	16.9	30.8	12.4	3.8	1.9
U3 - Toxicity prior Beta(1, 1)	36.4	18.1	28.6	11.9	3.1	1.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.6	6.1	6.6	3.6	1.8	
U3 - Toxicity prior Beta(0.15, 0.85)	11.6	6.2	6.6	3.5	1.8	
U2 - Toxicity prior Beta(0.3, 0.7)	11.1	5.5	6.7	4.2	2.1	
U3 - Toxicity prior Beta(0.3, 0.7)	11.1	5.5	6.8	4.2	2.1	
U2 - Toxicity prior Beta(0.5, 0.5)	10.8	5.1	6.7	4.7	2.3	
U3 - Toxicity prior Beta(0.5, 0.5)	10.9	5.1	6.7	4.7	2.3	
U2 - Toxicity prior Beta(1, 1)	10.7	5.2	6.5	4.8	2.5	
U3 - Toxicity prior Beta(1, 1)	10.7	5.2	6.5	4.8	2.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	2.5	1.2	0.8	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	2.5	1.2	0.8	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.3	1.2	0.8	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.3	1.2	0.8	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.3	1.2	0.8	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.2	1.1	0.8	0.2	
U2 - Toxicity prior Beta(1, 1)	0.3	2.2	1	0.8	0.2	
U3 - Toxicity prior Beta(1, 1)	0.3	2.2	1	0.8	0.2	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	19.9	40.8	20.6	7.1	1.4	10.2
U3 - Toxicity prior Beta(0.15, 0.85)	21.2	42.3	18.5	6.9	0.9	10.2
U2 - Toxicity prior Beta(0.3, 0.7)	18.7	39.4	20	10.3	1.8	9.8
U3 - Toxicity prior Beta(0.3, 0.7)	20.8	41.9	17	9.3	1.2	9.8
U2 - Toxicity prior Beta(0.5, 0.5)	18.5	38.2	21.2	10.3	2.3	9.5
U3 - Toxicity prior Beta(0.5, 0.5)	20.5	41.4	18.3	8.5	1.8	9.5
U2 - Toxicity prior Beta(1, 1)	19	36.2	21.5	11.2	2.5	9.6
U3 - Toxicity prior Beta(1, 1)	21.4	39.5	18.9	9	1.6	9.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.8	8.1	5.9	3.4	1.7	
U3 - Toxicity prior Beta(0.15, 0.85)	9.9	8.1	5.9	3.4	1.8	
U2 - Toxicity prior Beta(0.3, 0.7)	9.5	7.2	6.1	4.1	2.1	
U3 - Toxicity prior Beta(0.3, 0.7)	9.5	7.3	6.1	4.1	2	
U2 - Toxicity prior Beta(0.5, 0.5)	9.2	7	6	4.4	2.3	
U3 - Toxicity prior Beta(0.5, 0.5)	9.2	7.2	6	4.4	2.3	
U2 - Toxicity prior Beta(1, 1)	9.2	7.1	5.7	4.6	2.4	
U3 - Toxicity prior Beta(1, 1)	9.2	7.2	5.7	4.6	2.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3	1.1	0.9	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3	1.1	0.9	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.1	1	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.1	0.9	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.1	0.9	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.1	0.9	0.3	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1	0.9	0.2	
U3 - Toxicity prior Beta(1, 1)	0.1	2.6	1	0.9	0.2	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9	11.1	22.7	41	14.4	1.8
U3 - Toxicity prior Beta(0.15, 0.85)	14.6	10.3	20.5	41.5	11.3	1.8
U2 - Toxicity prior Beta(0.3, 0.7)	8.2	8.3	21.1	44.8	15.8	1.8
U3 - Toxicity prior Beta(0.3, 0.7)	12.6	9.3	20	43.3	13	1.8
U2 - Toxicity prior Beta(0.5, 0.5)	8	8.2	19.8	44.6	17.6	1.8
U3 - Toxicity prior Beta(0.5, 0.5)	12.4	9.1	17.5	44.3	14.9	1.8
U2 - Toxicity prior Beta(1, 1)	8	7.3	18.3	44.9	19.7	1.8
U3 - Toxicity prior Beta(1, 1)	11.4	8.7	16.8	44.6	16.7	1.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.6	4.7	6.5	8.2	5.8	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	4.6	6.6	8.1	5.9	
U2 - Toxicity prior Beta(0.3, 0.7)	4.3	4.3	6.2	8.7	6.3	
U3 - Toxicity prior Beta(0.3, 0.7)	4.3	4.3	6.2	8.7	6.3	
U2 - Toxicity prior Beta(0.5, 0.5)	4.2	4.1	6	8.8	6.7	
U3 - Toxicity prior Beta(0.5, 0.5)	4.3	4.1	5.9	8.8	6.6	
U2 - Toxicity prior Beta(1, 1)	4.2	4	5.8	8.6	7.1	
U3 - Toxicity prior Beta(1, 1)	4.2	4	5.9	8.6	7.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	1.4	2.3	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	1.4	2.3	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.8	1.4	2.2	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.7	1.4	2.2	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.3	2.2	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.3	2.2	0.7	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	1.3	2.1	0.7	
U3 - Toxicity prior Beta(1, 1)	0.2	2.7	1.3	2.1	0.7	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	5.5	31.6	27	22.4	9.3	4.2
U3 - Toxicity prior Beta(0.15, 0.85)	9	37.4	25.9	17.5	6	4.2
U2 - Toxicity prior Beta(0.3, 0.7)	5.4	31.4	26.2	22.4	10.4	4.2
U3 - Toxicity prior Beta(0.3, 0.7)	8	35	26.1	20.5	6.2	4.2
U2 - Toxicity prior Beta(0.5, 0.5)	5.1	30.1	26.1	23.1	11.4	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	8.2	33.6	26	20.9	7.1	4.2
U2 - Toxicity prior Beta(1, 1)	5.6	30.1	25.1	23.6	11.4	4.2
U3 - Toxicity prior Beta(1, 1)	7.5	35.9	24.5	20.8	7.1	4.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.9	6.5	6.5	6.1	5.4	
U3 - Toxicity prior Beta(0.15, 0.85)	5	6.4	6.6	6.1	5.4	
U2 - Toxicity prior Beta(0.3, 0.7)	4.8	6	6.5	6.5	5.8	
U3 - Toxicity prior Beta(0.3, 0.7)	4.8	6	6.4	6.5	5.8	
U2 - Toxicity prior Beta(0.5, 0.5)	4.7	5.9	6.3	6.6	6	
U3 - Toxicity prior Beta(0.5, 0.5)	4.8	5.9	6.3	6.6	6	
U2 - Toxicity prior Beta(1, 1)	4.7	5.8	6.3	6.5	6.3	
U3 - Toxicity prior Beta(1, 1)	4.8	5.8	6.2	6.5	6.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3.3	1.6	2.2	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.3	1.6	2.2	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3.2	1.7	2.2	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.2	1.6	2.3	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	3.2	1.6	2.2	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	3.2	1.6	2.2	1.1	
U2 - Toxicity prior Beta(1, 1)	0.2	3.1	1.6	2.2	1	
U3 - Toxicity prior Beta(1, 1)	0.2	3.2	1.6	2.2	1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	70	11.9	2.9	0.7	0.2	14.3
U3 - Toxicity prior Beta(0.15, 0.85)	70.4	11.6	2.5	0.9	0.3	14.3
U2 - Toxicity prior Beta(0.3, 0.7)	66.5	14	4.3	1	0.3	13.9
U3 - Toxicity prior Beta(0.3, 0.7)	69	13.1	2.8	1.1	0.2	13.8
U2 - Toxicity prior Beta(0.5, 0.5)	66.6	15	4	1	0.3	13.1
U3 - Toxicity prior Beta(0.5, 0.5)	68.4	13.6	3.3	1	0.4	13.3
U2 - Toxicity prior Beta(1, 1)	66.5	14.6	4.6	0.9	0.3	13.1
U3 - Toxicity prior Beta(1, 1)	68	14.1	3.5	0.8	0.3	13.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	19.6	4.3	2.2	0.9	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	19.5	4.3	2.2	1	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	18.9	4.4	2.6	1.2	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	18.8	4.5	2.6	1.2	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	18.6	4.5	2.6	1.3	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	18.6	4.4	2.7	1.4	0.6	
U2 - Toxicity prior Beta(1, 1)	18.4	4.6	2.7	1.3	0.6	
U3 - Toxicity prior Beta(1, 1)	18.5	4.5	2.6	1.3	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	1.5	0.4	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	1.6	0.4	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	1.4	0.4	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	1.5	0.4	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	1.3	0.3	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	1.3	0.3	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.3	1.2	0.3	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.2	1.2	0.3	0.2	0	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	27	37.8	19.3	10.4	3	2.5
U3 - Toxicity prior Beta(0.15, 0.85)	28.8	41.4	16	9.3	2.1	2.4
U2 - Toxicity prior Beta(0.3, 0.7)	26.6	35.2	19.4	12.8	3.7	2.3
U3 - Toxicity prior Beta(0.3, 0.7)	28.3	39.4	17	10.5	2.5	2.3
U2 - Toxicity prior Beta(0.5, 0.5)	25.4	35.7	19.3	12.6	4.7	2.3
U3 - Toxicity prior Beta(0.5, 0.5)	26.8	38.7	17.9	11.2	3.1	2.3
U2 - Toxicity prior Beta(1, 1)	25.6	36	19	12.9	4.2	2.3
U3 - Toxicity prior Beta(1, 1)	27.1	38.4	18.3	11.2	2.7	2.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.9	7.5	5.5	3.5	2.2	
U3 - Toxicity prior Beta(0.15, 0.85)	10.8	7.5	5.5	3.5	2.3	
U2 - Toxicity prior Beta(0.3, 0.7)	10.4	7	5.6	4.1	2.7	
U3 - Toxicity prior Beta(0.3, 0.7)	10.4	7.1	5.5	4	2.6	
U2 - Toxicity prior Beta(0.5, 0.5)	10.1	6.8	5.5	4.4	2.8	
U3 - Toxicity prior Beta(0.5, 0.5)	10.2	6.8	5.5	4.4	2.8	
U2 - Toxicity prior Beta(1, 1)	10.2	6.9	5.4	4.3	2.9	
U3 - Toxicity prior Beta(1, 1)	10.1	6.9	5.4	4.3	3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1	0.9	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1	0.9	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1	0.9	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1	0.9	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1	0.9	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1	0.9	0.3	
U2 - Toxicity prior Beta(1, 1)	0.2	2.4	0.9	0.8	0.3	
U3 - Toxicity prior Beta(1, 1)	0.1	2.4	0.9	0.8	0.3	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	44	8.2	2.4	0.6	0.4	44.4
U3 - Toxicity prior Beta(0.15, 0.85)	45.2	8	2	0.6	0.1	44.1
U2 - Toxicity prior Beta(0.3, 0.7)	44.2	8.6	3.3	1.1	0.5	42.3
U3 - Toxicity prior Beta(0.3, 0.7)	46	8.1	2	0.8	0.2	42.9
U2 - Toxicity prior Beta(0.5, 0.5)	43.5	9.1	3.2	1.2	0.6	42.4
U3 - Toxicity prior Beta(0.5, 0.5)	45.7	8.6	2.2	0.9	0.3	42.3
U2 - Toxicity prior Beta(1, 1)	42.3	9.8	3.3	1.7	0.4	42.5
U3 - Toxicity prior Beta(1, 1)	45	8.5	2.9	1.1	0.3	42.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.3	3.9	1.9	0.6	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	17.2	3.9	1.9	0.6	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	16.7	4.2	2.2	0.8	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	16.8	4.1	2.2	0.8	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	16.6	3.8	2.3	1.1	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	16.6	3.8	2.4	1	0.4	
U2 - Toxicity prior Beta(1, 1)	16.5	4	2.3	1	0.4	
U3 - Toxicity prior Beta(1, 1)	16.6	3.9	2.3	1	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	1.6	0.3	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	1.6	0.3	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	1.4	0.3	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	1.4	0.3	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	1.3	0.3	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	1.3	0.3	0.1	0	
U2 - Toxicity prior Beta(1, 1)	0.2	1.2	0.2	0.1	0	
U3 - Toxicity prior Beta(1, 1)	0.2	1.2	0.2	0.1	0	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.2	13.5	18.3	10.5	21	18.5
U3 - Toxicity prior Beta(0.15, 0.85)	23.6	14.4	14.8	9.9	18.6	18.7
U2 - Toxicity prior Beta(0.3, 0.7)	16.2	12.2	16.6	11.9	25.1	18
U3 - Toxicity prior Beta(0.3, 0.7)	21.8	13.4	14.3	10.5	21.9	18.1
U2 - Toxicity prior Beta(0.5, 0.5)	15.4	10	16.8	12.9	26.8	18.1
U3 - Toxicity prior Beta(0.5, 0.5)	20.1	12.3	14.9	11	23.5	18.2
U2 - Toxicity prior Beta(1, 1)	14	8.3	15.8	14.4	29.6	17.9
U3 - Toxicity prior Beta(1, 1)	18.6	12.9	12.9	12	25.5	18.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9	5	5.3	3.8	4.7	
U3 - Toxicity prior Beta(0.15, 0.85)	9	5.2	5.2	3.8	4.7	
U2 - Toxicity prior Beta(0.3, 0.7)	8.6	4.5	5.2	4.3	5.3	
U3 - Toxicity prior Beta(0.3, 0.7)	8.5	4.5	5.2	4.4	5.3	
U2 - Toxicity prior Beta(0.5, 0.5)	8.4	4.2	5	4.6	5.7	
U3 - Toxicity prior Beta(0.5, 0.5)	8.3	4.2	5.1	4.7	5.7	
U2 - Toxicity prior Beta(1, 1)	8.3	4.1	4.7	4.7	6.2	
U3 - Toxicity prior Beta(1, 1)	8.2	4.2	4.7	4.7	6.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	2.7	1.3	1.4	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	2.8	1.3	1.4	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.5	1.3	1.4	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.5	1.3	1.4	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.4	1.2	1.3	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.4	1.2	1.3	0.6	
U2 - Toxicity prior Beta(1, 1)	0.3	2.3	1	1.3	0.5	
U3 - Toxicity prior Beta(1, 1)	0.3	2.3	1	1.3	0.6	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14.3	30.7	10.7	4.9	1.7	37.7
U3 - Toxicity prior Beta(0.15, 0.85)	15.9	31.3	9.5	4.1	1.5	37.7
U2 - Toxicity prior Beta(0.3, 0.7)	13.8	29.5	12	6.3	1.7	36.7
U3 - Toxicity prior Beta(0.3, 0.7)	14.9	30.3	11.1	5.2	1.7	36.8
U2 - Toxicity prior Beta(0.5, 0.5)	12.2	29.3	13.2	6.6	2.2	36.5
U3 - Toxicity prior Beta(0.5, 0.5)	12.5	31.8	12.3	5.2	1.8	36.4
U2 - Toxicity prior Beta(1, 1)	12.3	29.9	12	6.6	2.7	36.5
U3 - Toxicity prior Beta(1, 1)	13.5	30.6	11.8	6	1.6	36.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.9	6.5	3.9	2	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	11.9	6.4	3.9	2	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	11.3	6.1	4.4	2.4	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	11.3	6.1	4.4	2.4	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	11	5.9	4.4	2.8	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	11	5.9	4.4	2.7	1.5	
U2 - Toxicity prior Beta(1, 1)	10.9	6.2	4.1	2.8	1.7	
U3 - Toxicity prior Beta(1, 1)	10.8	6.2	4.1	2.8	1.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.3	0.7	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.3	0.7	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.2	0.7	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.1	0.7	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2	0.7	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2	0.7	0.5	0.1	
U2 - Toxicity prior Beta(1, 1)	0.1	1.9	0.5	0.4	0.1	
U3 - Toxicity prior Beta(1, 1)	0.1	1.9	0.5	0.4	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	69.7	19.6	3.4	0.2	0	7.1
U3 - Toxicity prior Beta(0.15, 0.85)	70.3	20.4	1.7	0.3	0	7.3
U2 - Toxicity prior Beta(0.3, 0.7)	70.2	18.9	3.5	0.7	0	6.7
U3 - Toxicity prior Beta(0.3, 0.7)	72.3	17.6	2.5	0.9	0	6.7
U2 - Toxicity prior Beta(0.5, 0.5)	68.8	19.7	4	0.9	0	6.6
U3 - Toxicity prior Beta(0.5, 0.5)	71	18.7	3.5	0.5	0	6.3
U2 - Toxicity prior Beta(1, 1)	69.3	19.3	4	1	0	6.4
U3 - Toxicity prior Beta(1, 1)	71.3	18.4	3	0.7	0	6.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	18.6	6.3	2.9	0.9	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	18.4	6.4	3.1	0.9	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	17.6	6.3	3.6	1.2	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	17.6	6.3	3.6	1.2	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	17.5	6.2	3.7	1.3	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	17.5	6.2	3.7	1.4	0.2	
U2 - Toxicity prior Beta(1, 1)	17.3	6.3	3.6	1.4	0.3	
U3 - Toxicity prior Beta(1, 1)	17.4	6.2	3.6	1.4	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	2.4	0.6	0.3	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	2.4	0.6	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	2.2	0.6	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	2.2	0.6	0.3	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	2.1	0.5	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	2.1	0.5	0.3	0	
U2 - Toxicity prior Beta(1, 1)	0.5	1.8	0.4	0.3	0	
U3 - Toxicity prior Beta(1, 1)	0.6	1.8	0.4	0.2	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	16.2	16.7	10.8	35.2	17.9	3.2
U3 - Toxicity prior Beta(0.15, 0.85)	22.3	16.4	9.1	32.8	16.2	3.2
U2 - Toxicity prior Beta(0.3, 0.7)	16.9	13.6	10.8	35.5	20	3.2
U3 - Toxicity prior Beta(0.3, 0.7)	21.6	15.1	8.5	34	17.6	3.2
U2 - Toxicity prior Beta(0.5, 0.5)	14.3	13.4	10.4	37.5	21.2	3.2
U3 - Toxicity prior Beta(0.5, 0.5)	20.3	14.2	7.6	35.4	19.2	3.3
U2 - Toxicity prior Beta(1, 1)	14.7	12	9.8	36.6	23.6	3.3
U3 - Toxicity prior Beta(1, 1)	19.6	14.4	7.3	33.7	21.8	3.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.1	5.3	5.5	7.4	6.2	
U3 - Toxicity prior Beta(0.15, 0.85)	5.2	5.1	5.6	7.4	6.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4.9	4.8	5.3	7.9	6.8	
U3 - Toxicity prior Beta(0.3, 0.7)	4.8	4.8	5.3	7.9	6.8	
U2 - Toxicity prior Beta(0.5, 0.5)	4.8	4.4	5.1	8.2	7.1	
U3 - Toxicity prior Beta(0.5, 0.5)	4.8	4.4	5	8.2	7.1	
U2 - Toxicity prior Beta(1, 1)	4.7	4.3	4.8	7.8	7.9	
U3 - Toxicity prior Beta(1, 1)	4.7	4.3	4.9	7.8	7.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3	1.4	2.3	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.9	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	1.3	2.3	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	1.3	2.2	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	1.3	2.2	0.8	
U2 - Toxicity prior Beta(1, 1)	0.3	2.8	1.2	2.1	0.7	
U3 - Toxicity prior Beta(1, 1)	0.3	2.8	1.3	2.1	0.7	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	17.4	31	12.7	4.5	1.4	33
U3 - Toxicity prior Beta(0.15, 0.85)	19.1	32.2	10.1	4.6	0.9	33.1
U2 - Toxicity prior Beta(0.3, 0.7)	16	30.9	13.1	5.8	2.1	32.1
U3 - Toxicity prior Beta(0.3, 0.7)	17.5	32.3	11.1	5.4	1.5	32.2
U2 - Toxicity prior Beta(0.5, 0.5)	15	30.9	14.2	6	2.5	31.4
U3 - Toxicity prior Beta(0.5, 0.5)	17.2	32.5	12	4.8	2.1	31.4
U2 - Toxicity prior Beta(1, 1)	15	29.3	14.9	6.5	2.4	31.9
U3 - Toxicity prior Beta(1, 1)	16.5	31.5	12.7	5.7	1.8	31.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.5	6.7	4.3	2.3	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	11.6	6.7	4.2	2.2	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	10.8	6.5	4.6	2.7	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	10.8	6.4	4.6	2.8	1.5	
U2 - Toxicity prior Beta(0.5, 0.5)	10.5	6.1	4.7	3.2	1.7	
U3 - Toxicity prior Beta(0.5, 0.5)	10.6	6	4.7	3.2	1.7	
U2 - Toxicity prior Beta(1, 1)	10.5	6.3	4.5	3.2	1.8	
U3 - Toxicity prior Beta(1, 1)	10.4	6.3	4.6	3.2	1.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.6	0.8	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	0.8	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.4	0.8	0.6	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.4	0.8	0.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.3	0.7	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.3	0.7	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.1	0.6	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.1	0.6	0.6	0.1	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	24.5	17.2	15.6	22.5	12.8	7.4
U3 - Toxicity prior Beta(0.15, 0.85)	34.6	17.9	10.6	19.7	9.8	7.4
U2 - Toxicity prior Beta(0.3, 0.7)	22.4	16.5	13.8	25.6	14.5	7.2
U3 - Toxicity prior Beta(0.3, 0.7)	31	18	9.6	22.5	11.8	7.1
U2 - Toxicity prior Beta(0.5, 0.5)	20.4	16.5	13.5	26.8	15.7	7.1
U3 - Toxicity prior Beta(0.5, 0.5)	28.9	16.9	10.4	23.9	12.6	7.3
U2 - Toxicity prior Beta(1, 1)	20	14.3	12.5	28.8	17.2	7.2
U3 - Toxicity prior Beta(1, 1)	26.8	16.8	10.1	25.8	13.3	7.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.1	5.8	5.7	5.8	4.6	
U3 - Toxicity prior Beta(0.15, 0.85)	7.1	5.9	5.7	5.8	4.6	
U2 - Toxicity prior Beta(0.3, 0.7)	6.7	5.2	5.5	6.6	5.1	
U3 - Toxicity prior Beta(0.3, 0.7)	6.7	5.2	5.5	6.6	5.2	
U2 - Toxicity prior Beta(0.5, 0.5)	6.4	5	5.4	6.8	5.5	
U3 - Toxicity prior Beta(0.5, 0.5)	6.4	4.9	5.4	6.9	5.5	
U2 - Toxicity prior Beta(1, 1)	6.4	4.8	5.1	6.8	6	
U3 - Toxicity prior Beta(1, 1)	6.4	4.7	5.1	6.8	6.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	3.2	1.4	1.9	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.3	1.4	1.9	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3	1.5	2	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3	1.4	2	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3	1.4	1.9	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.9	1.4	1.9	0.7	
U2 - Toxicity prior Beta(1, 1)	0.5	2.8	1.3	1.8	0.6	
U3 - Toxicity prior Beta(1, 1)	0.5	2.8	1.3	1.8	0.6	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	50.8	28.5	9	3.4	0.9	7.4
U3 - Toxicity prior Beta(0.15, 0.85)	58.9	25.7	5.8	1.9	0.2	7.5
U2 - Toxicity prior Beta(0.3, 0.7)	49.5	28.7	9.7	3.7	1	7.4
U3 - Toxicity prior Beta(0.3, 0.7)	59.4	24.9	6	1.7	0.6	7.4
U2 - Toxicity prior Beta(0.5, 0.5)	48.5	29.2	10.4	3.3	1.2	7.4
U3 - Toxicity prior Beta(0.5, 0.5)	58.9	26.7	5	1.6	0.4	7.4
U2 - Toxicity prior Beta(1, 1)	49.3	27.7	10.3	4.3	1.1	7.3
U3 - Toxicity prior Beta(1, 1)	60.1	25.1	5.4	1.6	0.4	7.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.5	7.7	5.4	3	1.6	
U3 - Toxicity prior Beta(0.15, 0.85)	11.4	7.7	5.4	3.1	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	11.1	7.2	5.6	3.6	1.7	
U3 - Toxicity prior Beta(0.3, 0.7)	10.9	7.3	5.7	3.6	1.7	
U2 - Toxicity prior Beta(0.5, 0.5)	10.8	6.9	5.6	3.9	1.9	
U3 - Toxicity prior Beta(0.5, 0.5)	10.7	7	5.6	4	1.9	
U2 - Toxicity prior Beta(1, 1)	10.6	6.9	5.4	4.1	2.2	
U3 - Toxicity prior Beta(1, 1)	10.7	6.9	5.4	4.1	2.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	3.5	1.4	1.2	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	1.2	3.5	1.4	1.2	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	1.2	3.3	1.4	1.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	3.3	1.4	1.2	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	1	3.2	1.3	1.2	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	1	3.2	1.3	1.2	0.3	
U2 - Toxicity prior Beta(1, 1)	1	3	1.1	1.1	0.3	
U3 - Toxicity prior Beta(1, 1)	1	3	1.1	1.1	0.3	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	27.1	28.8	15.8	18.1	10.1	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	34	33.4	14.7	13.1	4.7	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	27.7	27.9	15.5	18.3	10.5	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	35.3	31.1	13.9	14.3	5.3	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	28.5	28.2	15.1	18	10.1	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	35.2	31.4	15.5	13	4.8	0.1
U2 - Toxicity prior Beta(1, 1)	28.1	28.6	14.7	17.5	11	0.1
U3 - Toxicity prior Beta(1, 1)	34.4	31.9	14.5	13.8	5.3	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.9	5.9	5.3	5.9	7	
U3 - Toxicity prior Beta(0.15, 0.85)	5.8	6	5.3	5.8	7	
U2 - Toxicity prior Beta(0.3, 0.7)	5.9	5.9	5.3	5.9	7.1	
U3 - Toxicity prior Beta(0.3, 0.7)	5.8	5.9	5.3	5.9	7.1	
U2 - Toxicity prior Beta(0.5, 0.5)	5.8	5.9	5.3	5.9	7.2	
U3 - Toxicity prior Beta(0.5, 0.5)	5.8	5.8	5.3	5.9	7.2	
U2 - Toxicity prior Beta(1, 1)	5.7	5.9	5.2	5.9	7.3	
U3 - Toxicity prior Beta(1, 1)	5.7	5.8	5.3	5.9	7.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	3.3	1.8	2.7	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	3.3	1.8	2.7	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	1	3.3	1.8	2.7	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	1	3.3	1.8	2.7	1.5	
U2 - Toxicity prior Beta(0.5, 0.5)	1	3.3	1.7	2.7	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	1	3.3	1.8	2.7	1.5	
U2 - Toxicity prior Beta(1, 1)	1.1	3.3	1.7	2.6	1.4	
U3 - Toxicity prior Beta(1, 1)	1	3.3	1.8	2.6	1.4	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	32.8	51	15.1	0.8	0	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	35.1	50.5	13.3	0.8	0	0.3
U2 - Toxicity prior Beta(0.3, 0.7)	31	51.6	15.6	1.5	0	0.3
U3 - Toxicity prior Beta(0.3, 0.7)	35.2	49.9	13.4	1.2	0	0.3
U2 - Toxicity prior Beta(0.5, 0.5)	31.1	50.5	16.1	2	0	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	32.5	52.7	12.6	1.9	0	0.3
U2 - Toxicity prior Beta(1, 1)	29.9	52	15.8	2	0	0.3
U3 - Toxicity prior Beta(1, 1)	32.7	53	12.4	1.6	0	0.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.4	10.4	6.5	2.2	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	10.3	10.3	6.8	2.1	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	9.8	10.1	6.9	2.7	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	9.8	10.1	6.9	2.7	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	9.5	10.1	6.9	2.9	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	9.5	10.2	6.9	2.9	0.5	
U2 - Toxicity prior Beta(1, 1)	9.3	10.2	6.8	3	0.6	
U3 - Toxicity prior Beta(1, 1)	9.3	10.3	6.7	3	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	2.9	1.1	0.5	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	2.9	1.1	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	2.7	1	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	2.8	1	0.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.6	1	0.5	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.6	0.9	0.5	0	
U2 - Toxicity prior Beta(1, 1)	0.4	2.5	0.8	0.5	0	
U3 - Toxicity prior Beta(1, 1)	0.4	2.5	0.8	0.5	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	10	11.9	12.8	17.7	46.9	0.7
U3 - Toxicity prior Beta(0.15, 0.85)	16.8	14.8	11.1	16	40.6	0.7
U2 - Toxicity prior Beta(0.3, 0.7)	9.7	11	11.5	17.1	50	0.7
U3 - Toxicity prior Beta(0.3, 0.7)	15.7	14.3	10.4	15.9	43	0.7
U2 - Toxicity prior Beta(0.5, 0.5)	9.4	10.3	10.3	16.6	52.7	0.7
U3 - Toxicity prior Beta(0.5, 0.5)	15.7	13.6	10.8	14.6	44.6	0.7
U2 - Toxicity prior Beta(1, 1)	8.6	8.8	10.5	15.3	56.1	0.7
U3 - Toxicity prior Beta(1, 1)	14.3	12.2	10	14.8	48	0.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.8	4	4.9	5.9	11.3	
U3 - Toxicity prior Beta(0.15, 0.85)	3.8	4	4.8	5.9	11.4	
U2 - Toxicity prior Beta(0.3, 0.7)	3.8	3.9	4.7	5.9	11.6	
U3 - Toxicity prior Beta(0.3, 0.7)	3.7	3.9	4.8	5.9	11.6	
U2 - Toxicity prior Beta(0.5, 0.5)	3.8	3.9	4.6	5.9	11.8	
U3 - Toxicity prior Beta(0.5, 0.5)	3.8	3.8	4.6	5.8	11.8	
U2 - Toxicity prior Beta(1, 1)	3.8	3.8	4.5	5.6	12.2	
U3 - Toxicity prior Beta(1, 1)	3.8	3.8	4.5	5.7	12.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.4	2.4	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.8	1.4	2.4	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.4	2.4	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.4	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.4	2.4	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.4	1.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	1.3	2.3	1.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.8	1.3	2.3	1.4	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	25.6	49.7	14.7	1.6	0	8.4
U3 - Toxicity prior Beta(0.15, 0.85)	28.9	48.7	12.4	1.6	0	8.4
U2 - Toxicity prior Beta(0.3, 0.7)	24.9	48.1	16.5	2.1	0	8.4
U3 - Toxicity prior Beta(0.3, 0.7)	26.6	49.8	13.4	1.7	0	8.5
U2 - Toxicity prior Beta(0.5, 0.5)	24.9	48.1	16.8	1.9	0.1	8.2
U3 - Toxicity prior Beta(0.5, 0.5)	27.1	48.8	14	1.7	0.1	8.3
U2 - Toxicity prior Beta(1, 1)	23.4	49.9	16.5	1.8	0.1	8.3
U3 - Toxicity prior Beta(1, 1)	25.6	49.9	14.4	1.6	0.1	8.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.7	10.4	6.5	2.1	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	9.8	10.4	6.4	2.1	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	9	10.1	7	2.6	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	9	10.2	6.9	2.6	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	8.9	9.9	6.9	3	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	8.8	9.7	7.1	3	0.5	
U2 - Toxicity prior Beta(1, 1)	8.6	9.7	7	3.2	0.6	
U3 - Toxicity prior Beta(1, 1)	8.7	9.7	7	3.2	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.6	1.1	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.6	1.1	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.4	1.1	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.4	1.1	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1	0.7	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1	0.7	0	
U2 - Toxicity prior Beta(1, 1)	0.4	2.9	0.9	0.6	0	
U3 - Toxicity prior Beta(1, 1)	0.4	2.9	0.9	0.6	0	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14.9	46.2	27.4	6.3	0.9	4.3
U3 - Toxicity prior Beta(0.15, 0.85)	20.3	48	22.7	4.3	0.4	4.3
U2 - Toxicity prior Beta(0.3, 0.7)	16.9	44.7	24.3	8.7	1.2	4.2
U3 - Toxicity prior Beta(0.3, 0.7)	20.5	47	21.2	6.5	0.6	4.2
U2 - Toxicity prior Beta(0.5, 0.5)	15	44.6	25.7	8.8	1.7	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	20.1	45.9	22.5	6.4	0.9	4.2
U2 - Toxicity prior Beta(1, 1)	15	42.4	26.8	9.9	1.7	4.2
U3 - Toxicity prior Beta(1, 1)	18.5	45.8	22.7	8	0.8	4.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.9	9	8.2	4.4	1.9	
U3 - Toxicity prior Beta(0.15, 0.85)	5.9	9	8.2	4.4	1.9	
U2 - Toxicity prior Beta(0.3, 0.7)	5.7	8.3	8.1	5.3	2.2	
U3 - Toxicity prior Beta(0.3, 0.7)	5.8	8.3	8	5.2	2.2	
U2 - Toxicity prior Beta(0.5, 0.5)	5.6	8	7.9	5.5	2.4	
U3 - Toxicity prior Beta(0.5, 0.5)	5.6	8	7.9	5.6	2.4	
U2 - Toxicity prior Beta(1, 1)	5.5	7.7	7.7	5.8	2.7	
U3 - Toxicity prior Beta(1, 1)	5.4	7.9	7.7	5.8	2.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.6	1.8	1.4	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.6	1.7	1.4	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.5	1.7	1.5	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.6	1.7	1.5	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.4	1.6	1.5	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.4	1.7	1.5	0.4	
U2 - Toxicity prior Beta(1, 1)	0.4	3.3	1.5	1.4	0.3	
U3 - Toxicity prior Beta(1, 1)	0.4	3.4	1.5	1.4	0.4	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	57.9	1.5	0	0	0	40.6
U3 - Toxicity prior Beta(0.15, 0.85)	57.8	1.3	0	0	0	40.9
U2 - Toxicity prior Beta(0.3, 0.7)	59.2	1.6	0	0	0	39.2
U3 - Toxicity prior Beta(0.3, 0.7)	59	1.5	0	0	0	39.5
U2 - Toxicity prior Beta(0.5, 0.5)	59.6	1.2	0.1	0	0	39.1
U3 - Toxicity prior Beta(0.5, 0.5)	59.6	1.2	0	0	0	39.2
U2 - Toxicity prior Beta(1, 1)	59.8	1.3	0.1	0	0	38.8
U3 - Toxicity prior Beta(1, 1)	59.7	1.4	0.1	0	0	38.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	20.5	2.2	0.6	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	20.6	2.2	0.6	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	20.4	2.4	0.6	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	20.5	2.4	0.6	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	20.4	2.5	0.6	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	20.4	2.5	0.6	0.1	0	
U2 - Toxicity prior Beta(1, 1)	20.4	2.5	0.6	0.1	0	
U3 - Toxicity prior Beta(1, 1)	20.3	2.5	0.6	0.1	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	0.6	0.1	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	0.6	0.1	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	0.6	0.1	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	0.6	0.1	0	0	
U2 - Toxicity prior Beta(1, 1)	0	0.6	0.1	0	0	
U3 - Toxicity prior Beta(1, 1)	0	0.6	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	29.2	52.6	12.1	1.3	0.1	4.7
U3 - Toxicity prior Beta(0.15, 0.85)	33.1	49.6	11.1	1.6	0	4.6
U2 - Toxicity prior Beta(0.3, 0.7)	31.2	50.3	12.2	1.4	0	4.9
U3 - Toxicity prior Beta(0.3, 0.7)	34.3	49.7	10.2	1	0	4.8
U2 - Toxicity prior Beta(0.5, 0.5)	31.1	49.2	13.7	1.6	0	4.4
U3 - Toxicity prior Beta(0.5, 0.5)	35	47.8	11.2	1.7	0	4.3
U2 - Toxicity prior Beta(1, 1)	30.6	50.7	12.1	2.4	0	4.2
U3 - Toxicity prior Beta(1, 1)	34.2	49.7	9.5	2.4	0	4.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.7	11.9	5.8	1.7	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	9.7	12	5.8	1.6	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	9.1	11.6	6.2	2.3	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	9.2	11.5	6.2	2.3	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	9	11	6.4	2.7	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	8.9	11.2	6.3	2.7	0.4	
U2 - Toxicity prior Beta(1, 1)	8.6	11.1	6.5	2.8	0.5	
U3 - Toxicity prior Beta(1, 1)	8.7	11	6.6	2.8	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	3.9	1.1	0.5	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	3.9	1.1	0.5	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	3.6	1.1	0.5	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	3.6	1.1	0.6	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	3.4	1	0.5	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	3.4	1	0.5	0	
U2 - Toxicity prior Beta(1, 1)	0.6	3.1	0.8	0.5	0	
U3 - Toxicity prior Beta(1, 1)	0.7	3.1	0.8	0.5	0	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	20.3	4.2	0.4	0	0	75.1
U3 - Toxicity prior Beta(0.15, 0.85)	20.7	4.3	0.2	0	0	74.8
U2 - Toxicity prior Beta(0.3, 0.7)	20.7	3	0.6	0	0	75.7
U3 - Toxicity prior Beta(0.3, 0.7)	21.1	2.9	0.1	0	0	75.9
U2 - Toxicity prior Beta(0.5, 0.5)	20.3	4.2	0.5	0	0	75
U3 - Toxicity prior Beta(0.5, 0.5)	20.5	4	0.2	0	0	75.3
U2 - Toxicity prior Beta(1, 1)	20.4	3.6	0.4	0	0	75.6
U3 - Toxicity prior Beta(1, 1)	21.1	3.6	0	0	0	75.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	16.7	3	0.9	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	16.6	2.9	0.9	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	16.6	3.2	1	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	16.6	3.2	1	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	16.6	3.1	1	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	16.7	3.1	1	0.1	0	
U2 - Toxicity prior Beta(1, 1)	16.6	3.2	1	0.1	0	
U3 - Toxicity prior Beta(1, 1)	16.6	3.3	1	0.1	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	1.3	0.2	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	1.3	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	1.2	0.1	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	1.3	0.1	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	1.2	0.1	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	1.2	0.1	0	0	
U2 - Toxicity prior Beta(1, 1)	0.1	1.1	0.1	0	0	
U3 - Toxicity prior Beta(1, 1)	0.1	1.1	0.1	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	1.7	6	47	44.6	0
U3 - Toxicity prior Beta(0.15, 0.85)	2.3	2.7	8	48.9	38.1	0
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	1.7	6.6	46.9	44.1	0
U3 - Toxicity prior Beta(0.3, 0.7)	2.4	2.9	7	48.2	39.5	0
U2 - Toxicity prior Beta(0.5, 0.5)	0.9	1.7	6.3	47.5	43.6	0
U3 - Toxicity prior Beta(0.5, 0.5)	2.2	2.9	7.2	49.2	38.5	0
U2 - Toxicity prior Beta(1, 1)	0.7	1.6	6.2	47.1	44.4	0
U3 - Toxicity prior Beta(1, 1)	2.3	2.5	6.6	48.2	40.4	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.3	3.2	3.7	8.7	11.2	
U3 - Toxicity prior Beta(0.15, 0.85)	3.3	3.2	3.7	8.6	11.3	
U2 - Toxicity prior Beta(0.3, 0.7)	3.3	3.1	3.7	8.6	11.3	
U3 - Toxicity prior Beta(0.3, 0.7)	3.3	3.2	3.6	8.6	11.3	
U2 - Toxicity prior Beta(0.5, 0.5)	3.2	3.1	3.6	8.6	11.3	
U3 - Toxicity prior Beta(0.5, 0.5)	3.2	3.2	3.6	8.6	11.3	
U2 - Toxicity prior Beta(1, 1)	3.2	3.2	3.6	8.6	11.4	
U3 - Toxicity prior Beta(1, 1)	3.2	3.2	3.6	8.6	11.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.5	1.1	3.2	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.5	1.1	3.2	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.5	1.1	3.2	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.5	1.1	3.2	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1.1	3.2	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1.1	3.2	1.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.5	1.1	3.2	1.1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.5	1.1	3.1	1.2	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	35.5	33.6	27	3.2	0	0.7
U3 - Toxicity prior Beta(0.15, 0.85)	41.3	32.1	23.4	2.4	0.1	0.7
U2 - Toxicity prior Beta(0.3, 0.7)	35.9	31.2	28.7	3.4	0.1	0.7
U3 - Toxicity prior Beta(0.3, 0.7)	39.9	31.7	25.4	2.2	0.1	0.7
U2 - Toxicity prior Beta(0.5, 0.5)	34.3	31.7	29.1	4.2	0	0.7
U3 - Toxicity prior Beta(0.5, 0.5)	40.1	30.9	24.1	4.1	0.1	0.7
U2 - Toxicity prior Beta(1, 1)	36.1	30.5	28.2	4.4	0.1	0.7
U3 - Toxicity prior Beta(1, 1)	40.5	30.6	24.3	3.8	0.1	0.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.8	8.9	8.1	2.7	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	9.9	8.8	8.1	2.7	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	9.5	8.2	8.4	3.3	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	9.5	8.2	8.4	3.3	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	9.2	8.1	8.4	3.6	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	9.3	8.1	8.4	3.6	0.6	
U2 - Toxicity prior Beta(1, 1)	8.9	8.1	8.4	3.7	0.7	
U3 - Toxicity prior Beta(1, 1)	9	8	8.5	3.7	0.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	3.2	1.4	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	3.2	1.4	0.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	3	1.4	0.8	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	3	1.4	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	2.9	1.2	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	2.9	1.3	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.8	2.8	1.1	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	0.8	2.8	1.1	0.7	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	33.7	29	21	5.3	0.1	10.9
U3 - Toxicity prior Beta(0.15, 0.85)	37.2	29.3	18.2	4.2	0.2	10.9
U2 - Toxicity prior Beta(0.3, 0.7)	33.9	28.4	21.3	5.8	0.1	10.5
U3 - Toxicity prior Beta(0.3, 0.7)	37.7	28.5	18.8	4.5	0.1	10.4
U2 - Toxicity prior Beta(0.5, 0.5)	33.8	27	23.4	5.8	0	10
U3 - Toxicity prior Beta(0.5, 0.5)	37.2	27.7	19.9	5.2	0	10
U2 - Toxicity prior Beta(1, 1)	34.1	27.6	21.3	6.9	0.1	10
U3 - Toxicity prior Beta(1, 1)	36.9	27.3	19	6.7	0	10.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.3	7.6	6.9	2.6	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	11.3	7.6	6.9	2.7	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	10.8	7.2	7.2	3.2	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	10.8	7.1	7.3	3.2	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	10.6	6.8	7.4	3.6	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	10.6	6.8	7.5	3.5	0.6	
U2 - Toxicity prior Beta(1, 1)	10.3	6.9	7.3	3.8	0.7	
U3 - Toxicity prior Beta(1, 1)	10.4	6.8	7.4	3.8	0.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.1	1.1	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3	1.1	0.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	2.9	1.1	0.8	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	2.9	1.1	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.7	1	0.8	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.7	1	0.8	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	2.5	0.8	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	2.5	0.8	0.7	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	24.9	28	34.6	6.7	0.4	5.4
U3 - Toxicity prior Beta(0.15, 0.85)	28.1	29.2	30.5	6.6	0.2	5.4
U2 - Toxicity prior Beta(0.3, 0.7)	22.4	26.5	37.5	7.9	0.3	5.4
U3 - Toxicity prior Beta(0.3, 0.7)	26.5	26.2	33.8	7.6	0.4	5.5
U2 - Toxicity prior Beta(0.5, 0.5)	23.3	24.3	38.7	7.8	0.6	5.3
U3 - Toxicity prior Beta(0.5, 0.5)	26.8	25	34.5	7.7	0.7	5.3
U2 - Toxicity prior Beta(1, 1)	23.2	23.3	38.4	9.1	0.7	5.3
U3 - Toxicity prior Beta(1, 1)	27.3	25.7	32.7	8.4	0.6	5.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.4	7.1	8.9	3.8	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	8.3	7.4	8.7	3.9	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	8.1	6.7	8.8	4.5	1.3	
U3 - Toxicity prior Beta(0.3, 0.7)	8.1	6.7	8.8	4.6	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	8	6.2	8.7	5	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	8	6.2	8.7	5	1.6	
U2 - Toxicity prior Beta(1, 1)	7.8	6.2	8.5	5.2	1.8	
U3 - Toxicity prior Beta(1, 1)	7.8	6.3	8.4	5.2	1.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.1	1.6	1	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.1	1.5	1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	2.9	1.5	1	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	2.9	1.5	1	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.8	1.4	1	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.8	1.4	1	0.2	
U2 - Toxicity prior Beta(1, 1)	0.4	2.7	1.2	0.9	0.2	
U3 - Toxicity prior Beta(1, 1)	0.4	2.7	1.2	0.9	0.2	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	27.3	31.7	6	0.4	0	34.6
U3 - Toxicity prior Beta(0.15, 0.85)	27.2	34.3	4	0.2	0	34.3
U2 - Toxicity prior Beta(0.3, 0.7)	25.3	35.1	6.4	0.2	0	33
U3 - Toxicity prior Beta(0.3, 0.7)	26.8	35.2	5.1	0.2	0	32.7
U2 - Toxicity prior Beta(0.5, 0.5)	25.8	34.1	7	0.3	0	32.8
U3 - Toxicity prior Beta(0.5, 0.5)	27.1	34.1	5.2	0.6	0	33
U2 - Toxicity prior Beta(1, 1)	26.2	32.4	7.7	0.5	0	33.2
U3 - Toxicity prior Beta(1, 1)	27.3	33	5.7	0.6	0	33.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	12.9	8.3	3.8	0.8	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	12.9	8.6	3.7	0.8	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	12.1	8.7	4.2	1.1	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	12.1	8.6	4.2	1.1	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	11.9	8.3	4.5	1.4	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	11.9	8.3	4.5	1.4	0.2	
U2 - Toxicity prior Beta(1, 1)	11.6	8.4	4.6	1.4	0.2	
U3 - Toxicity prior Beta(1, 1)	11.6	8.5	4.6	1.4	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.7	0.6	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.7	0.6	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.5	0.6	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.5	0.6	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.4	0.6	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.4	0.6	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.2	2.1	0.5	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.2	2.1	0.5	0.2	0	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	70.2	23.1	3.7	0.2	0	2.8
U3 - Toxicity prior Beta(0.15, 0.85)	71.1	23.2	2.9	0.1	0	2.7
U2 - Toxicity prior Beta(0.3, 0.7)	68.6	25.4	3.2	0.1	0	2.7
U3 - Toxicity prior Beta(0.3, 0.7)	70.8	23.7	2.7	0.1	0	2.7
U2 - Toxicity prior Beta(0.5, 0.5)	68.3	25	3.8	0.2	0	2.7
U3 - Toxicity prior Beta(0.5, 0.5)	69.4	24.2	3.6	0.2	0	2.6
U2 - Toxicity prior Beta(1, 1)	68.9	24.7	3.7	0.2	0	2.5
U3 - Toxicity prior Beta(1, 1)	70	24.1	3.3	0.1	0	2.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	18.2	7.6	3.1	0.5	0	
U3 - Toxicity prior Beta(0.15, 0.85)	18.2	7.6	3.1	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	17.6	7.8	3.4	0.7	0	
U3 - Toxicity prior Beta(0.3, 0.7)	17.6	7.7	3.4	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	17.1	8	3.6	0.8	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	17.1	7.9	3.6	0.8	0.1	
U2 - Toxicity prior Beta(1, 1)	17	8.1	3.6	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	17	8	3.6	0.8	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2	0.7	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	2.1	0.6	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	1.9	0.6	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	1.9	0.6	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	1.8	0.6	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	1.8	0.5	0.1	0	
U2 - Toxicity prior Beta(1, 1)	0.5	1.6	0.5	0.1	0	
U3 - Toxicity prior Beta(1, 1)	0.5	1.6	0.5	0.1	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	19.1	19.9	16.4	34.4	9.4	0.8
U3 - Toxicity prior Beta(0.15, 0.85)	25.9	23.3	12.9	29.5	7.6	0.8
U2 - Toxicity prior Beta(0.3, 0.7)	18.4	18	15.8	37.7	9.3	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	24.4	20.9	13.6	33.6	6.7	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	17.4	18.3	15.7	36.2	11.6	0.8
U3 - Toxicity prior Beta(0.5, 0.5)	24.3	21.8	12.7	32	8.4	0.8
U2 - Toxicity prior Beta(1, 1)	19.1	16.1	14.8	36.5	12.7	0.8
U3 - Toxicity prior Beta(1, 1)	25	20.6	11.4	32.3	9.9	0.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.2	5.1	6.1	8.3	5.3	
U3 - Toxicity prior Beta(0.15, 0.85)	5.1	5.1	6.1	8.4	5.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4.9	4.8	5.9	8.6	5.7	
U3 - Toxicity prior Beta(0.3, 0.7)	4.9	4.8	5.8	8.6	5.7	
U2 - Toxicity prior Beta(0.5, 0.5)	4.9	4.6	5.7	8.7	6.1	
U3 - Toxicity prior Beta(0.5, 0.5)	4.9	4.6	5.7	8.7	6	
U2 - Toxicity prior Beta(1, 1)	4.9	4.4	5.5	8.6	6.5	
U3 - Toxicity prior Beta(1, 1)	4.9	4.5	5.5	8.5	6.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.9	1.6	2.3	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.9	1.6	2.3	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.8	1.6	2.3	0.9	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.8	1.6	2.3	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.8	1.5	2.2	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	1.5	2.2	0.8	
U2 - Toxicity prior Beta(1, 1)	0.4	2.8	1.4	2.1	0.8	
U3 - Toxicity prior Beta(1, 1)	0.3	2.8	1.4	2.1	0.8	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	5.7	32.1	30.2	24.2	7.7	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	7.5	38.4	27.9	20.9	5.2	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	5.3	32.2	29.1	24.8	8.5	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	6.6	37.9	28	22	5.4	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	5.8	33.1	30.2	23	7.8	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	6.5	38.9	27.6	21.2	5.7	0.1
U2 - Toxicity prior Beta(1, 1)	6	31.6	29.3	25.3	7.7	0.1
U3 - Toxicity prior Beta(1, 1)	7.9	39.8	26.6	21.7	3.9	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4	5.7	6.5	8	5.6	
U3 - Toxicity prior Beta(0.15, 0.85)	4.1	5.8	6.5	8	5.6	
U2 - Toxicity prior Beta(0.3, 0.7)	4.1	5.6	6.4	8.2	5.8	
U3 - Toxicity prior Beta(0.3, 0.7)	4	5.6	6.4	8.2	5.8	
U2 - Toxicity prior Beta(0.5, 0.5)	4	5.7	6.3	8	6	
U3 - Toxicity prior Beta(0.5, 0.5)	4.1	5.6	6.2	8.1	6	
U2 - Toxicity prior Beta(1, 1)	4	5.6	6.1	8.1	6.2	
U3 - Toxicity prior Beta(1, 1)	4	5.7	6.1	8	6.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	1.4	1.9	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	1.4	1.8	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.8	1.4	1.8	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.8	1.4	1.8	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.8	1.4	1.8	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.3	1.8	1.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.7	1.3	1.8	1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.8	1.3	1.7	1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.9	16.5	32.6	29.9	10.5	0.6
U3 - Toxicity prior Beta(0.15, 0.85)	15	20.2	30	28	6.2	0.6
U2 - Toxicity prior Beta(0.3, 0.7)	9.5	17.3	31	30	11.6	0.6
U3 - Toxicity prior Beta(0.3, 0.7)	14.2	22	30.4	25.6	7.2	0.6
U2 - Toxicity prior Beta(0.5, 0.5)	9.4	15.5	31.9	30.3	12.3	0.6
U3 - Toxicity prior Beta(0.5, 0.5)	14.5	20.8	30.5	25.9	7.7	0.6
U2 - Toxicity prior Beta(1, 1)	9.2	14.4	31.2	30.3	14.3	0.6
U3 - Toxicity prior Beta(1, 1)	13.9	20.5	30.8	26	8.2	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4	4.4	7.5	7.6	6.4	
U3 - Toxicity prior Beta(0.15, 0.85)	4	4.5	7.4	7.7	6.4	
U2 - Toxicity prior Beta(0.3, 0.7)	3.9	4.3	7.3	7.8	6.6	
U3 - Toxicity prior Beta(0.3, 0.7)	3.9	4.3	7.2	7.8	6.7	
U2 - Toxicity prior Beta(0.5, 0.5)	3.9	4.2	7.1	7.9	6.8	
U3 - Toxicity prior Beta(0.5, 0.5)	3.9	4.2	7.1	7.8	6.9	
U2 - Toxicity prior Beta(1, 1)	3.9	4.2	6.9	7.9	7.1	
U3 - Toxicity prior Beta(1, 1)	3.9	4.2	7	7.8	7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	1.9	2.8	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	1.9	2.8	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	1.9	2.7	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	1.9	2.7	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	1.8	2.7	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	1.9	2.7	1.1	
U2 - Toxicity prior Beta(1, 1)	0.3	2.8	1.8	2.6	1.1	
U3 - Toxicity prior Beta(1, 1)	0.3	2.9	1.8	2.6	1.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	10.1	6.8	13.2	43.4	26.5	0
U3 - Toxicity prior Beta(0.15, 0.85)	14	10	12.6	39.9	23.5	0
U2 - Toxicity prior Beta(0.3, 0.7)	8.8	5.5	13.5	44.3	27.9	0
U3 - Toxicity prior Beta(0.3, 0.7)	13	8.4	13.2	40.8	24.6	0
U2 - Toxicity prior Beta(0.5, 0.5)	8.4	5.6	12.7	43.6	29.7	0
U3 - Toxicity prior Beta(0.5, 0.5)	12.4	8	12.4	41.8	25.4	0
U2 - Toxicity prior Beta(1, 1)	8.6	5.3	13.2	42.2	30.7	0
U3 - Toxicity prior Beta(1, 1)	11.9	8.4	12.2	40	27.5	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.5	3.6	5.2	8.7	8.9	
U3 - Toxicity prior Beta(0.15, 0.85)	3.6	3.6	5.2	8.7	8.9	
U2 - Toxicity prior Beta(0.3, 0.7)	3.5	3.6	5.1	8.7	9.2	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	3.5	5.1	8.7	9.2	
U2 - Toxicity prior Beta(0.5, 0.5)	3.5	3.5	5	8.6	9.4	
U3 - Toxicity prior Beta(0.5, 0.5)	3.5	3.5	5	8.6	9.4	
U2 - Toxicity prior Beta(1, 1)	3.5	3.5	4.9	8.4	9.8	
U3 - Toxicity prior Beta(1, 1)	3.5	3.5	4.9	8.4	9.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.8	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.3	2.8	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.8	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.8	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.8	1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.8	1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.3	2.7	0.9	
U3 - Toxicity prior Beta(1, 1)	0.2	2.6	1.3	2.7	0.9	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.3	17.5	44	14.9	3.5	1.8
U3 - Toxicity prior Beta(0.15, 0.85)	24.4	19.1	38.9	13.5	2.3	1.8
U2 - Toxicity prior Beta(0.3, 0.7)	18.1	17.2	41.6	16.1	5.2	1.8
U3 - Toxicity prior Beta(0.3, 0.7)	23.6	17.1	37.6	16.6	3.1	2
U2 - Toxicity prior Beta(0.5, 0.5)	16	16.3	40.4	19.3	6.2	1.8
U3 - Toxicity prior Beta(0.5, 0.5)	21.8	16.6	38.2	17.6	3.9	1.9
U2 - Toxicity prior Beta(1, 1)	16.1	15.7	40.3	19.5	6.5	1.9
U3 - Toxicity prior Beta(1, 1)	22.1	17.8	36.9	16.9	4.5	1.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.5	6.5	9.3	5.4	3	
U3 - Toxicity prior Beta(0.15, 0.85)	5.6	6.4	9.2	5.4	3	
U2 - Toxicity prior Beta(0.3, 0.7)	5.3	5.9	8.8	6.3	3.5	
U3 - Toxicity prior Beta(0.3, 0.7)	5.3	5.9	8.8	6.3	3.4	
U2 - Toxicity prior Beta(0.5, 0.5)	5	5.6	8.8	6.6	3.7	
U3 - Toxicity prior Beta(0.5, 0.5)	5.1	5.6	8.8	6.6	3.7	
U2 - Toxicity prior Beta(1, 1)	5	5.4	8.6	6.6	4.1	
U3 - Toxicity prior Beta(1, 1)	5	5.4	8.5	6.6	4.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.3	1.9	1.6	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.3	1.9	1.6	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.1	1.9	1.7	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.1	1.9	1.7	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3	1.8	1.6	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3	1.8	1.6	0.6	
U2 - Toxicity prior Beta(1, 1)	0.5	3	1.7	1.6	0.5	
U3 - Toxicity prior Beta(1, 1)	0.5	3	1.7	1.6	0.5	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	11.9	41.2	33.9	11.4	1.5	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	15.5	46	28.8	8.4	1.2	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	11.2	40.9	32.7	13.1	2	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	14.9	44.3	27.7	11.5	1.5	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	11.4	40.1	32.6	13.4	2.4	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	15.1	44.9	27.1	10.9	1.9	0.1
U2 - Toxicity prior Beta(1, 1)	11.2	39.9	32.3	13.5	3	0.1
U3 - Toxicity prior Beta(1, 1)	14.8	45.2	26.9	10.8	2.2	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.6	8.7	9	5.2	2.5	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	8.6	9.1	5.2	2.5	
U2 - Toxicity prior Beta(0.3, 0.7)	4.5	8.2	8.7	5.7	2.8	
U3 - Toxicity prior Beta(0.3, 0.7)	4.5	8.3	8.7	5.8	2.8	
U2 - Toxicity prior Beta(0.5, 0.5)	4.5	8.1	8.4	6	3	
U3 - Toxicity prior Beta(0.5, 0.5)	4.5	8.1	8.4	5.9	3	
U2 - Toxicity prior Beta(1, 1)	4.4	7.9	8.3	6.1	3.3	
U3 - Toxicity prior Beta(1, 1)	4.4	8	8.2	6.1	3.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3.1	1.6	1.4	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3.1	1.6	1.4	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	3.1	1.6	1.4	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	3	1.6	1.4	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3	1.6	1.3	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	3.1	1.6	1.3	0.5	
U2 - Toxicity prior Beta(1, 1)	0.3	3	1.5	1.3	0.4	
U3 - Toxicity prior Beta(1, 1)	0.3	3	1.4	1.3	0.4	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	24	40.6	12.3	5.9	2.1	15.1
U3 - Toxicity prior Beta(0.15, 0.85)	29.5	37.5	11.2	4.5	1.6	15.7
U2 - Toxicity prior Beta(0.3, 0.7)	23	38	14.1	7.2	2.9	14.8
U3 - Toxicity prior Beta(0.3, 0.7)	27	38.6	11.4	6.3	2.1	14.6
U2 - Toxicity prior Beta(0.5, 0.5)	22.8	37.3	15	7.3	3.4	14.2
U3 - Toxicity prior Beta(0.5, 0.5)	27.3	37	12.9	5.8	2.5	14.5
U2 - Toxicity prior Beta(1, 1)	21.2	37.5	14.3	9.6	3	14.4
U3 - Toxicity prior Beta(1, 1)	26.9	38.1	11.4	6.9	1.9	14.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.8	8.8	5.2	2.8	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	9.9	8.8	5.3	2.7	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	9.1	8.4	5.6	3.4	1.8	
U3 - Toxicity prior Beta(0.3, 0.7)	9	8.4	5.6	3.4	1.8	
U2 - Toxicity prior Beta(0.5, 0.5)	8.8	7.8	5.9	3.9	1.9	
U3 - Toxicity prior Beta(0.5, 0.5)	8.8	7.8	5.8	3.9	2	
U2 - Toxicity prior Beta(1, 1)	8.6	8.1	5.5	4	2.2	
U3 - Toxicity prior Beta(1, 1)	8.7	7.9	5.6	4	2.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.5	1	0.9	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.5	1	0.8	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.2	1.1	0.9	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.2	1	0.9	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3	1	0.9	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3	1	0.8	0.2	
U2 - Toxicity prior Beta(1, 1)	0.3	2.8	0.8	0.8	0.2	
U3 - Toxicity prior Beta(1, 1)	0.4	2.8	0.8	0.8	0.2	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	28.9	47.6	13.6	2.1	0.4	7.4
U3 - Toxicity prior Beta(0.15, 0.85)	36.4	44.9	9.5	1.6	0.4	7.2
U2 - Toxicity prior Beta(0.3, 0.7)	28.3	47.5	13.7	3.1	0.7	6.7
U3 - Toxicity prior Beta(0.3, 0.7)	34.4	45.7	10	2.6	0.4	6.9
U2 - Toxicity prior Beta(0.5, 0.5)	30	44.8	14.7	3.2	1	6.3
U3 - Toxicity prior Beta(0.5, 0.5)	35.1	44.6	10.5	2.8	0.6	6.4
U2 - Toxicity prior Beta(1, 1)	29.9	44.5	14.8	3.5	0.9	6.4
U3 - Toxicity prior Beta(1, 1)	35	44.1	11.3	2.8	0.5	6.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.1	10.4	6.1	2.3	1	
U3 - Toxicity prior Beta(0.15, 0.85)	9.2	10.3	6.1	2.3	1	
U2 - Toxicity prior Beta(0.3, 0.7)	8.8	9.9	6.2	2.9	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	8.6	10.1	6.2	2.9	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	8.3	9.6	6.5	3.3	1.3	
U3 - Toxicity prior Beta(0.5, 0.5)	8.2	9.7	6.4	3.4	1.3	
U2 - Toxicity prior Beta(1, 1)	8.2	9.6	6.2	3.6	1.5	
U3 - Toxicity prior Beta(1, 1)	8.2	9.7	6.2	3.6	1.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	4	1.2	0.8	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	4	1.2	0.8	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	3.7	1.1	0.9	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.8	1.1	0.8	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.5	1.1	0.8	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.5	1.1	0.9	0.1	
U2 - Toxicity prior Beta(1, 1)	0.6	3.2	0.9	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	0.6	3.2	0.9	0.8	0.1	